GRADUATE SCHOOL SOLAR ACADEMY

UNIVERSITÉ SAVOIE MONT BLANC

cnrs

Operational research for urban solar development

"Regional PV production modelling from open data"



21/10 (TP 1) and 22/10/2026 (TP 2) Session 3

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Curriculum

Plan

Day	Time	Duration	Content
Session 1 Monday 19/10	8h00 - 9h30	1 h 30	Lecture 70 % / Hands-on 30 %
Session 2 Tuesday 20/10	13h15 - 14h45	1 h 30	Lecture 30 % / Hands-on 70 %
Session 3 **Wednesday 21/10 (TP 1)** **Thursday 22/10 (TP 2)**	**8h15 - 11h15 (TP 1)** **13h00 - 16h00 (TP 2)**	**3 h**	**Lecture 15 % / Hands-on 60 % / Project 25 %**
Session 4 Monday 09/11 (TP 1) Tuesday 10/11 (TP 2)	8h15 - 11h15 (TP 1) 8h15 - 11h15 (TP 2)	3 h	Lecture 10 % / Project 90 %
Session 5 Tuesday 24/11 (TP 2) Friday 27/11 (TP 1)	9h45 - 12h45 (TP 2) 13h15 - 16h15 (TP 1)	3 h	Project 100 %
Oral Wednesday 02/12	Afternoon, both TP groups	3 h	Oral presentations of part B, 10 min per pair
Total		15 h	

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Agenda

Review notebook 2

PV performance model steps 6 to 8

Notebook 3: one system, one year

Project launch: part A on your own system

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Review notebook 2

Notebook recap

Google Colab link: [link]

Correction: walked through in class

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Review notebook 2

Python commands

```
import matplotlib.pyplot as plt
fig, ax = plt.subplots(1, 1, figsize=(9, 5))  # a figure with one axis
df.plot(ax=ax)  # plot a DataFrame on the axis
fig, (ax1, ax2) = plt.subplots(2, 1, sharex=True)  # two axes sharing the x axis
filter_1 = weather["ghi"] > 800  # a boolean Series
filter_2 = weather.index.month == 6
weather.loc[filter_1 & filter_2, "dhi"]  # rows where both conditions hold
cos_aoi = cos_aoi.clip(lower=0)  # force values below 0 up to 0
assert abs(zenith - 22.64) < 0.05  # self-check: stops with an error if false
```

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Modeling steps

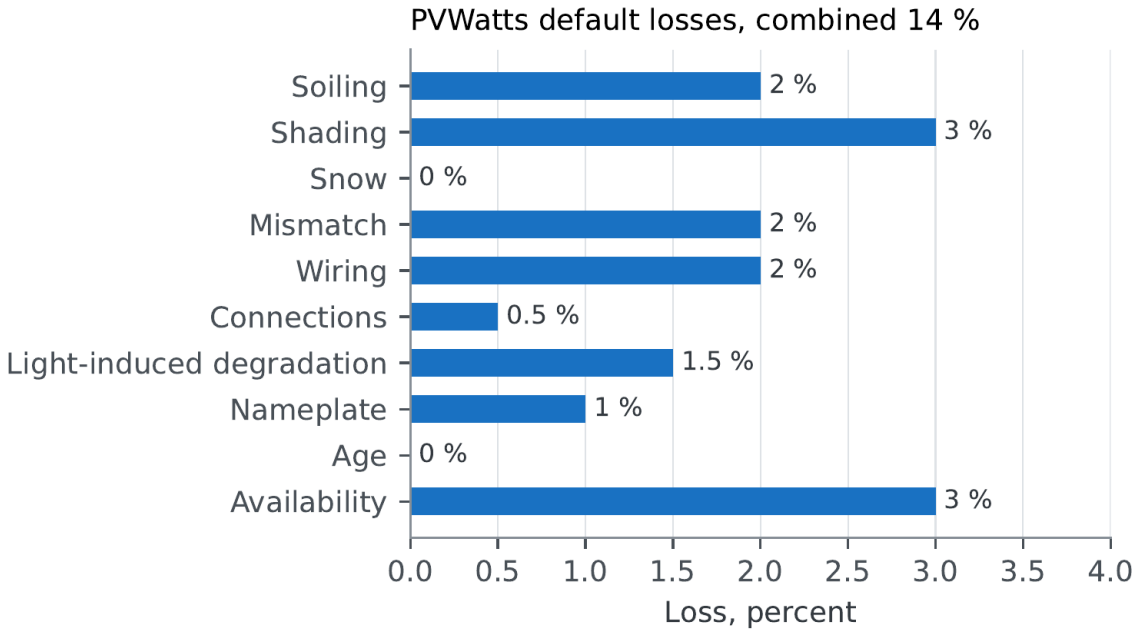
6. DC losses

Losses combine multiplicatively, not additively.

PVWatts defaults: soiling 2 %, shading 3 %, snow 0, mismatch 2 %, wiring 2 %, connections 0.5 %, light-induced degradation 1.5 %, nameplate 1 %, age 0, availability 3 %. Combined 14 %.

Age: 0.5 % per year is a common assumption. Snow: several percent in the Alps.

These are defaults, not measurements. State the ones you keep.



Dobos (2014), NREL/TP-6A20-62641.

Modeling steps

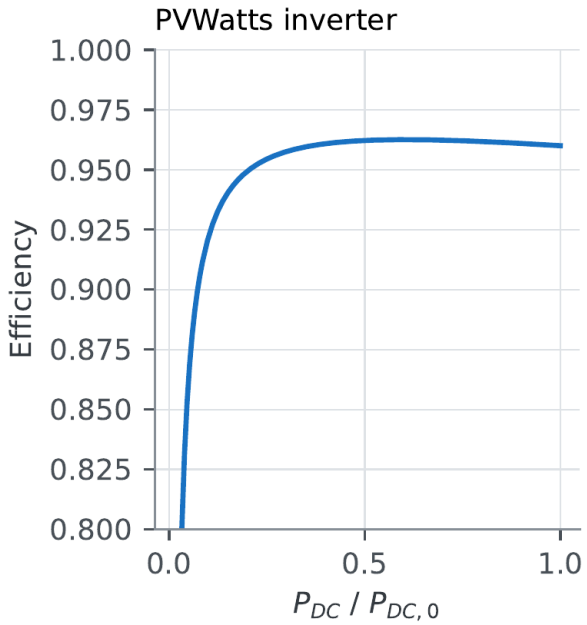
7. Inverter and AC power

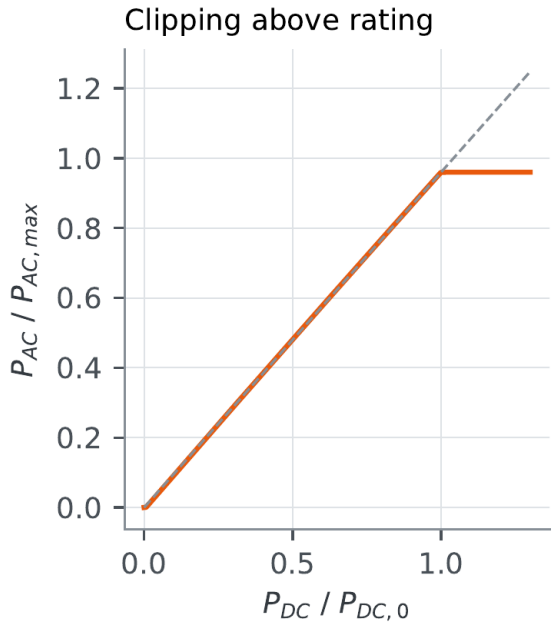
Peak efficiency 96 to 98 %, reached between 20 and 60 % of rated load. Falls below 10 % of rated load.

Loading ratio P0 / PAC,max between 1.0 and 1.3. Above 1.0 the inverter clips the highest hours.

Annual clipping loss near or below 1 % for a ratio of 1.2 in France.

PVWatts inverter model: efficiency as a function of the load, then clipping at the AC rating.





Dobos (2014). pvlib.inverter.pvwatts

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Modeling steps

8. Performance metrics

Energy: E = Σ PAC(t) · Δt, with Δt = 1 h.

Specific yield: Yf = E / P0 in kWh/kWp. 950 to 1100 in northern France, 1250 to 1450 in the south.

Performance ratio, IEC 61724-1: PR = E / (P0 · HPOA / Gref). Removes the resource, leaves the system quality. 0.78 to 0.85 for a recent rooftop.

Energy performance index: EPI = E measured / E modelled. Below 1 when the system under-performs the model.

A PR above 0.90 in a model: a missing loss. A yield of 1800 kWh/kWp in Savoie: a W to Wh error.

$$PR = \frac{E}{P_0 H_{POA} / G_{ref}}$$

IEC 61724-1. Ranges consistent with PVGIS outputs for French sites.

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Modeling steps

8. Error metrics against a reference

MBE: sign and size of the bias. MAE: typical absolute error. RMSE: weights the large hours.

Normalise by the mean reference power or by the installed capacity, and state which one. The two differ by a factor near 8.

Report at three time scales: hourly, daily, monthly. Bias survives aggregation. Random scatter shrinks with the averaging window.

R^2 on the hourly scatter and the slope of the fit complete the picture.

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{t}(m_t - r_t)^2}$$

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Notebook 3

AC power estimation

Google Colab link: [link]

I. Data visualization: weather and operating data of the course system (3 kWp, tilt 30, south).

II. POA irradiance, module temperature, DC power, AC power with the PVWatts inverter, each compared with the actuals.

III. PR, EPI, and the accuracy of the models: MBE, MAE, RMSE. Three outage days to find.

This notebook is part A of the project, on the course site. The project repeats it on your own system.

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Project launch: part A on your own system

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Project

Instructions

Part A, individual: one system per student, student number n. POA irradiance, module temperature, DC power, AC power, PR and EPI, compared with the actuals.

Part B, in pairs: one region per pair, pair number k. Register, a first model, a better representation of the fleet, comparison with RTE, error attribution.

Deliverables: one notebook per pair as html, the csv outputs with the naming convention, one oral presentation per pair on part B.

Deadline: Tuesday 01/12/2026 at 23h59. Oral: Wednesday 02/12, afternoon. Late: −10 % under 15 min, −50 % under 1 h, −100 % after.

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Project

Datasets

Per student n: student_n_inputs.csv (weather at the site), student_n_actuals.csv (g_poa, t_mod, p_dc, p_ac), student_n_system.csv.

Per region: register.csv, nodes.csv, weather_dept_<dd>.csv for every department, reference_2023.csv from RTE eCO2mix (EDF SEI for Corse), hourly, MW, UTC.

pairs.csv gives the region of each group: twelve regions, one each. Corse comes with the EDF SEI hourly series instead of RTE. Occitanie is the worked example of notebook 4.

Orientation survey: only for Occitanie. For your region, build the representation from what the comparison says, then state the assumption.

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Project

Grading

80 % notebook and csv files, 20 % oral. Part A is graded from the notebook and the csv files only. The oral covers part B.

- 30 % part A, individual: 6 % per task. Relevance of the method and RMSE accuracy: 3/5 for the basic method, 5/5 for the teacher's accuracy.
- 45 % part B, pair: 5 % for the data check B.0, then 8 % per task. The first model, the finding on the profile, the reasoning behind your classes, the metrics, one tested hypothesis.
- 5 % structure and ease of reviewing the notebook.
- 20 % oral on part B, pair: 10 min, then questions to each member.

Bonus 10 %: one weather node against one per department, with the change of the metrics.

Advice: refer to the course formulas, concise explanations, one plot per result, comments in the code.

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Project

Start now: part A

Open the project notebook: https://colab.research.google.com/github/Mohamed-ENNHIRI/ENER958/blob/main/notebooks/project.ipynb

First action: File, Save a copy in Drive. Then write your pair number k in the first cell. The notebook finds your members and your region.

Part A reuses notebook 3 on your own system: POA, module temperature, DC power, AC power, PR and EPI.

Today: A.1 and A.2, load your data and compute the POA irradiance and the module temperature. Compare with your actuals.

Part B, the regional fleet in pairs, starts in session 4 with notebook 4.

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That’s it

Next session: from one system to a region, part B of the project. Finish part A before.